

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Detailed Marking Report — Chemistry Exam-20

Seemab · Class 9 (Science Group) · Revision Test, Chapters 1 to 9 · 60-MCQ paper · Attempted 22 August 2026

59 / 71

TOTAL SCORE

83.1%

PERCENTAGE

A

GRADE

49 / 60

SECTION A
81.7%

6 / 6

SECTION B
100%

4 / 5

SECTION C
80%

How this paper was marked

Seemab again wrote her sixty MCQ answers as a numbered list on the first page rather than filling the bubble grid. Her Section A score of 49/60 sits below her lifetime baseline, so under the standing rule the whole paper was verified twice over before any mark was recorded. **First pass:** every letter re-read from the scan at 400 DPI and the answer key re-extracted independently from the published PDF rather than from a stored list. **Second pass:** all sixty correct answers were re-derived from the printed exam paper itself, reading the options off the rendered page and deciding each answer independently before comparing with the key. That found **zero key errors across all sixty questions** — which matters, because an error on one of the forty-nine she got right would have cost her a mark that nobody was checking for. The first 400 DPI pass also had a cropping fault: rows were cut using uniform spacing, but the page is warped from the scanner's de-skew, so several crops drifted onto neighbouring lines. It was redone with overlapping bands that keep each number beside its letter. **Q34 in particular is now unambiguous** — a cancelled A followed by B, and B is correct. A guard was also run against a systematic numbering slip: her list scores 49 aligned, and only 10 to 14 under any shift of one or two places, so the numbering is sound. **All eleven wrong answers are genuine, sit on stated Student Learning Outcomes, and needed no voiding.**

SECTION A — Multiple Choice (49 / 60)

#	Topic	Hers	Key	Mark	Comment and source in her textbook
1	Essential question → biochemistry	C	C	1	Correct. Ch1 §1.3, SLO 3.
2	Essential question → geochemistry	C	C	1	Correct. Ch1 §1.3, SLO 3.
3	Essential question → nuclear chemistry	D	D	1	Correct. Ch1 §1.3, SLO 3.
4	Astrochemistry defined	B	B	1	Correct. Ch1 §1.2.
5	Metal dissolved in mercury	B	B	1	Correct, an amalgam. Ch2 §2.4.
6	Fog, clouds and mist	A	A	1	Correct, the solvent is a gas. Ch2 §2.4.
7	Hydrogen on nickel, ghee industry	A	A	1	Correct, a solid solution. The solvent decides the type. Ch2 §2.4.
8	What solubility depends on	C	C	1	Correct, pressure. Ch2 §2.4.7.
9	Carbonated drink	B	B	1	Correct, a gas in a liquid. Ch2 §2.4.
10	What an atomic model is	B	B	1	Correct. Ch3 §3.1 SLO.
11	Why tritium is absent naturally	A	A	1	Correct, its nucleus is highly unstable. Ch3 §3.5.1.
12	Definition of a radioisotope	D	D	1	Correct. Ch3 §3.2.
13	Neutrons in ^{80}Br	D	D	1	Correct, $80 - 35 = 45$. Ch3 §3.3.
14	Proton emission during decay	A	B	0	She chose "another isotope of the same element". That is what emitting a NEUTRON does. Emitting a proton changes the proton number, so the atom becomes another element. Ch3 §3.3. Valid question.
15	What the nuclear force acts between	D	D	1	Correct, proton-proton, neutron-neutron and neutron-proton. Ch3 §3.1.
16	Electron deflection in a field	A	A	1	Correct, the electron's much smaller mass. Ch3 §3.2 SLO.
17	Rutherford's force balance	C	A	0	She chose the gravitational force. In Rutherford's planetary model the centrifugal force of the revolving electron balances the electrostatic attraction. Ch3 §3.1. Valid, though this is the most detailed model question on the paper.
18	Auf Bau order after 3p	B	B	1	Correct, 4s. Ch3 §3.7.
19	What favours anion formation	D	D	1	Correct, a high electron affinity. Ch3 §3.6.
20	What decides an element's block	B	B	1	Correct, the sub-shell taking the last electron. Ch4 §4.1.1.
21	Group of $1s^2 2s^2 2p^2$	B	C	0	She chose Group 4. She counted the four valence electrons correctly but did not add 10 for the p block. Group = $4 + 10 = 14$. Ch4 §4.1.1. A one-step slip, not a gap.

22	Group 16 collective name	C	C	1	Correct, chalcogens. Ch4 §4.2.
23	Density across a period	A	A	1	Correct, peaks near the middle. Ch4 §4.4.3.
24	Highest first ionization energy	D	D	1	Correct, Si at 786 kJ/mol. Ch4 §4.3.3.
25	Most thermally stable hydrogen halide	D	D	1	Correct, HF. Ch4 §4.7.
26	Why transition metals melt high	C	C	1	Correct, partially filled d sub-shells. Ch4 §4.5.
27	Form of the noble gases	D	D	1	Correct, mono-atomic. Ch4 §4.8.
28	Form of the halogens	C	C	1	Correct, diatomic. Ch4 §4.7.
29	Least reactive alkali metal	B	B	1	Correct, lithium. Ch4 §4.4.
30	What the period number equals	A	C	0	She chose the number of valence electrons, which is the GROUP rule for the s block. The period is the principal quantum number n of the valence shell. Ch4 §4.1.1. Pairs with Q21 as one confusion.
31	Atoms with 5, 6 or 7 valence electrons	B	B	1	Correct, they gain electrons and form anions. Ch4 §4.2.
32	Why non-metals melt low	A	A	1	Correct, weaker covalent, van der Waals or hydrogen bonds. Ch4 §4.9.
33	Why CO ₂ melts low	B	C	0	She chose "it contains free electrons". CO ₂ has strong bonds INSIDE each molecule but weak forces BETWEEN molecules. Ch5 §5.5. Free electrons belong to metals.
34	C-N bond in HCN	B	B	1	Correct, triple. She first wrote A, cancelled it and wrote B. The final answer is clear. Ch5 §5.3.2.
35	Why BF ₃ is electron deficient	B	B	1	Correct, only three pairs on boron. Ch5 §5.3.4.
36	Adhesive action of paints and dyes	C	C	1	Correct, hydrogen bonding. Ch5 Key Points.
37	What holds graphite's layers	A	A	1	Correct, weak van der Waals forces. Ch5 §5.5.1.
38	Configuration of Be ²⁺	C	C	1	Correct, helium. Ch5 §5.1 duplet rule.
39	Third factor fixing properties	D	D	1	Correct, the arrangement of the particles. Ch5 §5.5.
40	Why metals are malleable	C	C	1	Correct, layers of cations slide. Ch5 §5.6.1.
41	Ionic compounds in compression	C	C	1	Correct, brittle. Ch5 §5.5.3.
42	Ionic compounds in batteries	D	D	1	Correct, electrolytes. Ch5 §5.5.3.
43	Empirical = molecular formula	A	A	1	Correct, CH ₄ . Ch6 §6.1.
44	Formula mass of Ca(OH) ₂	B	B	1	Correct, 74. Ch6 §6.2.
45	Moles in 1.5 kg of MgO	D	D	1	Correct, 37.5 mol. She handled the kg→g conversion, which was new on this paper. Ch6 §6.5.1.
46	Atoms in 23 g of sodium	B	B	1	Correct, 6.022×10^{23} . Ch6 §6.4.
47	Balancing Ca + H ₂ O	C	C	1	Correct, 1, 2, 1, 1. Ch6 §6.6.1.
48	Ions taking no part in a reaction	D	D	1	Correct, spectator ions. Ch6 §6.6.2.
49	Molecular formula of CH ₃ -CH ₂ -NH ₂	A	A	1	Correct, C ₂ H ₇ N. Ch6 §6.7.
50	Formula from Ca ²⁺ and PO ₄ ³⁻	B	C	0	She chose Ca ₂ (PO ₄) ₃ , swapping the criss-cross. Balancing charge needs 3(+2) = 2(-3), so Ca ₃ (PO ₄) ₂ . Ch6 §6.3. Same slip family as exam-19 Q20.
51	Aluminium + nitrogen named	A	D	0	She chose aluminium NITRATE. Nitrate is the polyatomic NO ₃ ⁻ . A binary compound takes the -ide suffix, so aluminium nitride. Ch6 §6.3.
52	Mass of 1.5 mol of N ₂	A	A	1	Correct, 42 g. She used the diatomic molar mass 28, which was the trap. Ch6 §6.5.1.
53	Molecules in 0.5 mol of H ₂ O ₂	A	A	1	Correct, 3.011×10^{23} . Ch6 §6.5.2.
54	Oxidation number of N in NO	B	B	1	Correct, +2. Ch7 §7.2.2.
55	Reducing agent in CO ₂ + 2Mg	B	A	0	She chose CO ₂ , which is the OXIDIZING agent. Mg goes 0 → +2, so Mg is oxidized and is therefore the reducing agent. Ch7 §7.4.2. The two roles are inverted.
56	Sum of oxidation numbers in an ion	D	D	1	Correct, equals the charge on the ion. Ch7 §7.2.2.
57	What counts as the agent	B	A	0	She chose the single atom. The textbook states in bold that the agent is the whole molecule or formula unit containing that atom. Ch7 §7.4.2.
58	Ag ⁺ + e ⁻ → Ag	B	B	1	Correct, reduced because it gains an electron. Ch7 §7.1.3.

59	What an effective collision overcomes	B	D	0	She chose the container walls. The colliding particles must overcome the repulsion between their surrounding electrons. Ch8 §8.5.
60	What a catalyst does not change	C	A	0	She chose “mechanism of the reaction”, but the stem itself says the catalyst PROVIDES a new mechanism. What it cannot change is the overall enthalpy. Ch8 §8.6.

SECTION B — Short Questions (6 / 6)

Q	Task	Awarded	Comment
2(i)	30.4% N / 69.6% O, $M_r = 92 \rightarrow$ empirical then molecular formula	3 / 3	Full marks. She assumed 100 g, divided by the atomic masses to get N 2.17 and O 4.35, divided both by the smaller value to reach a clean 1 : 2, and wrote NO₂ . Then empirical mass $14 + 2(16) = 46$, $92 \div 46 = 2$, and multiplied up to N₂O₄ . Every step shown and labelled.
2(ii)	Dot-and-cross for NH ₃ ; count bond pairs and lone pairs	3 / 3	She scribbled out a first attempt and redrew it. The second diagram has nitrogen central with three overlapping hydrogen circles and a dot-and-cross pair at each bond. She then wrote “ Bond pairs: 3, Lone pairs: 1 ”, both correct. One read could not be made with certainty: the lone pair on nitrogen is not cleanly separable from the left-hand bonding pair, because the ruled lines of the page run straight through that region. The mark was awarded on the strength of her correct written count in part (b). If that diagram is read as missing the lone pair, this part becomes 2/3 and the paper 58/71.

SECTION C — Long Question (4 / 5)

Part	Task	Awarded	Comment
3(a)	Formula mass of K ₂ SO ₄	1 / 1	$2(39) + 32 + 4(16) = 78 + 32 + 64 = \mathbf{174 \text{ amu}}$. Correct, with the working set out.
3(b)	Moles in 60.9 g	1 / 1	She wrote the formula, corrected “formula mass” to “molar mass” herself, then $60.9 \div 174 = \mathbf{0.35 \text{ mol}}$. Correct.
3(c)	Number of formula units	1 / 1	$0.35 \times 6.022 \times 10^{23} = \mathbf{2.11 \times 10^{23}}$. Correct.
3(d)	Balance $\text{KOH} + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$	1 / 1	She wrote the unbalanced equation first, then underneath $2\text{KOH} + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$. Correct.
3(e)	Rewrite the balanced equation with state symbols	0 / 1	Not attempted. The page ends after part (d) with five blank ruled lines below it, and the scan has all four pages, so this is a skip and not a missing page. The answer needed was $2\text{KOH}(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{K}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$.

Where the marks actually went

Chapter	Score	Comment
Ch1 Nature of Chemistry	4 / 4	Clean, including all three of the essential-question items that had never been examined before.
Ch2 Matter	5 / 5	Clean.
Ch3 Atomic Structure	8 / 10	Q14 proton emission, Q17 Rutherford’s force balance.
Ch4 Periodic Table	11 / 13	Q21 and Q30, which are the same confusion twice, see below.
Ch5 Chemical Bonding	9 / 10	Only Q33. A clear recovery — this was the block that cost her four marks on exam-19.
Ch6 Stoichiometry	9 / 11	Q50 and Q51, both formula-and-naming rather than calculation. Every numerical was right.
Ch7 Electrochemistry	3 / 5	Q55 and Q57. The oxidizing/reducing inversion is the one to fix.
Ch8 Energetics	0 / 2	Q59 and Q60. See the caution below before reading anything into this.

1. Group number and period number are being mixed up

Q21 gave the configuration $1s^2 2s^2 2p^2$ and asked for the group. She counted four valence electrons correctly, then answered **Group 4** instead of adding 10 for the p block. Q30 asked what the period number equals and she answered **the number of valence electrons** — which is the group rule for the s block. These are the same confusion pointing in two directions, and it is worth ten minutes: **period = n of the valence shell; group = valence electrons, plus 10 in the p block.**

2. The oxidizing and reducing agent are inverted

Q55 gave $\text{CO}_2 + 2\text{Mg} \rightarrow 2\text{MgO} + \text{C}$ and asked for the reducing agent. She named CO_2 , which is the oxidizing agent. Q57 then asked what counts as the agent and she chose the single atom rather than the whole molecule. She scored 2/2 on Chapter 7 in exam-19, so this is a slip rather than a hole — but redox-agent identification has now been examined on six papers and is worth locking down. The rule is one line: **the species that is oxidized is the reducing agent.**

3. Answering less than the question asked, now on five consecutive papers

Section C(e) asked her to rewrite the balanced equation with state symbols. She balanced part (d) perfectly and then stopped. The same mark went the same way on Maths exam-15 (did not name the properties), Chemistry exam-19 (balanced perfectly, omitted the state symbols), and twice on Maths exam-16. **Every one of these marks was lost after the hard work was already done.** This is still the single highest-value habit available to her.

A caution on the Chapter 8 zero

Chapter 8 was worth two of the sixty marks and she got neither. That is not evidence of a chapter-level gap. Energetics is a seven-page chapter that has been examined on eight previous papers, and by this paper the only angles left unused were two fine-grained clauses: the electron-repulsion wording inside the definition of an effective collision, and the line that a catalyst does not change the reaction enthalpy. Both are printed in the book and sit on stated outcomes, so both marks stand — but two questions on the most obscure sentences in a nearly exhausted chapter measure very little. The same caveat applies, more weakly, to Chapter 7 at five questions.

What the 60-MCQ format does to the percentage

This is her first paper with sixty MCQs. Section A alone carries 60 of the 71 marks, so Section A is very nearly the whole score, and doubling the question count again from exam-19's twenty-four reaches sub-topics that shorter papers never sampled. Nine of the eleven losses are on material examined either once before or never. **Do not read 83.1% against a record built on 65-mark papers where Sections B and C carried 53 of the marks and she has been near-perfect in them all year.** On this paper she was perfect in Section B and dropped a single mark in Section C.

— End of Marking Report —